RETURNS AND GROWTH RATES ON ONE PAGE

Differences, ratios and logarithms

Several honest ways to say how fast a price grew, and why the arithmetic mean of returns flatters a fluctuating one.

THE RULE

Change, ratio, and the logarithm of the ratio

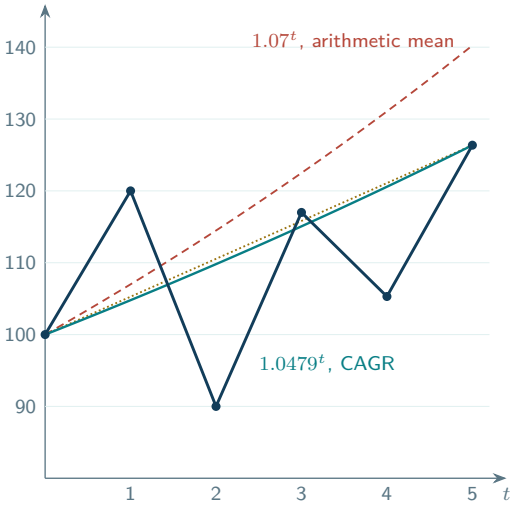
For prices $P_0, P_1, \ldots, P_T$ there are three ways to measure each step:

$$\Delta P_t = P_t - P_{t-1}, \qquad R_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \qquad r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t)$$

Each has its own natural way of combining over time:

$$\sum_t \Delta P_t = P_T - P_0, \qquad \prod_t (1 + R_t) = \frac{P_T}{P_0}, \qquad \sum_t r_t = \ln \frac{P_T}{P_0}.$$

First differences add but depend on the price level, simple returns multiply, and log returns add, which is why logs turn compounding back into ordinary sums; for small moves $r_t \approx R_t$.



The price path, the compound curve that fits its ends, and the straight line of first differences.

A FLUCTUATING PRICE

Five years of $+20\%$, -25%, $+30\%$, -10%, $+20\%$

year t	0	1	2	3	4	5	combined
price P_t	100	120	90	117	105.30	126.36	
first difference ΔP_t		$+20$	-30	$+27$	-11.70	$+21.06$	sum $+26.36$
simple return R_t		$+20\%$	-25%	$+30\%$	-10%	$+20\%$	product of $1 + R_t$: 1.2636
log return r_t		$+0.1823$	-0.2877	$+0.2624$	-0.1054	$+0.1823$	sum $0.2340 = \ln 1.2636$

The two $+20\%$ years have the same log return wherever they happen, while their first differences, $+20$ and $+21.06$, depend on the price they start from.

ANNUALISING

Five average rates for one price path

rate	formula	value	what it answers
arithmetic mean	$\frac{1}{T}\sum R_t$	7.00%	the expected one-year return, if each year is a fresh draw
simple average	$\frac{1}{T}\left(\frac{P_T}{P_0} - 1\right)$	5.27%	straight-line growth, the first differences spread evenly
geometric mean, CAGR	$\left(\frac{P_T}{P_0}\right)^{1/T} - 1$	4.79%	the steady yearly rate that reaches the same end price
mean log return	$\frac{1}{T}\sum r_t = \frac{1}{T}\ln\frac{P_T}{P_0}$	4.68%	the same growth as a continuously compounded rate, $\mathrm{e}^{0.0468} = 1.0479$
harmonic mean of $1 + R_t$	$\frac{T}{\sum 1/(1+R_t)} - 1$	2.45%	the lowest of the three classical means of the growth factors

Only the geometric mean and the mean log return reproduce the actual end price; $100 \times 1.07^5 = 140.26$ overshoots 126.36 badly. The order $7.00\% \geq 4.79\% \geq 2.45\%$ is the inequality AM $\geq$ GM $\geq$ HM applied to the growth factors $1 + R_t$.

VOLATILITY DRAG

Fluctuation costs growth

A gain of 50% followed by a loss of 50% has arithmetic mean 0 but leaves $1.5 \times 0.5 = 0.75$ of the start, a geometric mean of -13.4% a year. In general

$$\text{GM} \approx \text{AM} - \tfrac{1}{2}\sigma^2,$$

and here $\sigma = 20.9\%$ gives $7.00\% - 2.18\% = 4.82\%$, close to the true 4.79%. The wider the swings, the further the arithmetic mean runs ahead of what the price actually did.

WHERE THE HARMONIC MEAN EARNS ITS PLACE

Pound-cost averaging

Investing the same sum at each price from 100 to 105.30 buys more shares when the price is low, so the average cost per share is the harmonic mean of the prices, 105.29, below their arithmetic mean of 106.46. The harmonic mean belongs to prices bought with fixed money, just as it belonged to the sides of a box on the isoperimetric sheet.

Conventions. T is the number of years, σ the population standard deviation of the simple returns and CAGR the compound annual growth rate; the figures illustrate the arithmetic of returns and are not investment advice.